
Finding Variables Under Consideration: Probing LLM Pragmatic Defaults via Perplexity Sensitivity

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Abstract

When large language models help users write messages, they make implicit choices along pragmatic dimensions—formality, directness, politeness, emotional tone, and specificity—that shape the communicative impact of the resulting text. Yet it remains unclear which of these dimensions the model actively “considers” during generation versus which it can reason about only when explicitly prompted. We introduce *perplexity sensitivity analysis*, a behavioral probing method that detects pragmatic variables under consideration by measuring whether an LLM assigns significantly different likelihoods to response variants that differ along a single pragmatic axis. Across 30 message-writing scenarios spanning work-place, personal, and service domains, we find that two instruction-tuned models (QWEN2.5-7B-INSTRUCT and QWEN2.5-3B-INSTRUCT) exhibit strong, statistically significant directional preferences for four of five axes: polite over blunt ($d=1.81$), vague over specific ($d=1.02$), indirect over direct ($d=0.94$), and formal over casual ($d=0.68$). Emotional tone is the sole axis with no detectable preference. Comparing these behavioral results against model introspection reveals a striking dissociation: directness is strongly defaulted on but rarely self-reported, while emotional tone is frequently mentioned but not behaviorally detected. These results replicate perfectly across model sizes (Spearman $\rho=1.0$), suggesting that perplexity sensitivity analysis is a reliable, low-cost method for auditing pragmatic defaults in LLMs.

1 Introduction

When a user asks a language model to “write an email to my boss about missing the deadline,” the model must make a series of pragmatic choices: How formal should the tone be? Should the message hedge or be direct? How much detail is appropriate? These choices are rarely specified by the user, yet they shape whether the resulting message comes across as professional or casual, assertive or meek, empathetic or cold. In effect, every generated message encodes a set of implicit pragmatic defaults—but which variables does the model actually *consider* during generation, and which does it simply default on without deliberation?

The gap between knowledge and behavior. Prior work has shown that LLMs possess substantial pragmatic knowledge: they can identify scalar implicatures [Cho and Kim, 2024, Ruis et al., 2024], reason about speech acts when prompted [Sato et al., 2025], and explain pragmatic dimensions of text when asked [Lin et al., 2024]. However, knowing *about* a pragmatic variable is not the same as *considering* it during generation. Trienes et al. [2025] demonstrated that LLMs’ self-reported information salience diverges from their actual content selection behavior—models claim to prioritize certain content but behaviorally do otherwise. We hypothesize that a similar gap exists for pragmatic variables: models can discuss formality, directness, and tone when asked, but do not necessarily deliberate over these dimensions when producing text.

Our approach. We propose *perplexity sensitivity analysis*, a method that operationalizes “variables under consideration” through the lens of conditional log-likelihood. For a given message-writing prompt, we construct response pairs that differ along a single pragmatic axis (e.g., polite vs. blunt, direct vs. indirect) while holding other dimensions approximately constant. We then measure each variant’s log-likelihood under the model. If the model assigns systematically different likelihoods to the two poles of an axis across diverse scenarios, we conclude that the model has a *directional preference*—and thus that the axis is “under consideration” in the sense that the model’s generation prior encodes a default along it.

We apply this method to five pragmatic axes—formality, directness, politeness, emotional tone, and specificity—plus a synonym-substitution control, across 30 scenarios in three communicative domains (workplace, personal, service). We evaluate two models from the Qwen family (7B and 3B parameters) and compare behavioral detection against introspective probing of GPT-4.1.

Our main findings are:

- We find that LLMs exhibit a coherent pragmatic default profile—**polite, vague, indirect, and formal**—with effect sizes ranging from $d=0.68$ to $d=1.81$, while emotional tone shows no directional preference ($d=0.08$).
- We demonstrate a dissociation between behavioral defaults and introspective reports: directness is strongly defaulted on but rarely self-reported (6.7% introspection rate), while emotional tone is frequently mentioned (53.3%) but not behaviorally detected.
- We show perfect cross-model replication ($\rho=1.0$) across two model sizes, establishing perplexity sensitivity analysis as a reliable method for auditing LLM pragmatic defaults.

2 Related Work

Pragmatic competence in LLMs. A growing body of work evaluates whether LLMs understand pragmatic phenomena. Ruis et al. [2024] introduce the LUDWIG benchmark for conversational implicature and show that GPT-4 achieves 88.7% accuracy with chain-of-thought prompting, while smaller models perform near chance. Cho and Kim [2024] probe BERT and GPT-2 for scalar implicature inference and find that both models prefer pragmatic interpretations in neutral contexts, with GPT-2 showing increased surprisal when pragmatic inference is required under upper-bound questions. Lin et al. [2024] compare direct (embedding-based) and indirect (perplexity-based) methods for probing scalar adjective knowledge, finding that LLMs encode rich lexical-semantic information but show “unsatisfying performance on scalar diversity”—the pragmatic aspect. Sato et al. [2025] demonstrate that providing pragmatic theory descriptions (e.g., Grice’s maxims) as prompts improves LLM performance on implied-meaning tasks by up to 9.6%, suggesting that models possess latent pragmatic knowledge that is not always activated. Collectively, these studies establish that LLMs *can* reason about pragmatic phenomena but leave open whether they *do so* during unconstrained generation.

Perplexity and surprisal as probes. Using model-internal probability estimates to probe linguistic knowledge is well established. Lin et al. [2024] compare perplexity of correct vs. incorrect scalar constructions (e.g., “good but not awesome” vs. “awesome but not good”) to assess whether models encode intensity orderings. Cho and Kim [2024] measure surprisal under different question contexts to detect pragmatic processing. Cífka and Liutkus [2023] propose context length probing, which tracks how a model’s predictions change as context is incrementally extended, yielding importance scores via KL divergence. We build on this tradition but shift from probing *comprehension* to probing *generation preferences*: rather than asking whether the model assigns lower perplexity to linguistically correct constructions, we ask whether it shows systematic directional preferences along pragmatic axes.

Behavioral vs. introspective assessment. Trienes et al. [2025] use length-constrained summarization as a behavioral probe for information salience, tracking which content an LLM retains at different compression levels. They find that models’ behavioral salience hierarchies are consistent across model families but do *not* match their self-reported salience when asked directly. This behavioral-introspective gap motivates our comparison of perplexity-detected pragmatic preferences against introspective variable reports. Our work extends their finding from content salience to pragmatic variable consideration, using a complementary probing method (log-likelihood of pre-filled responses rather than constrained generation).

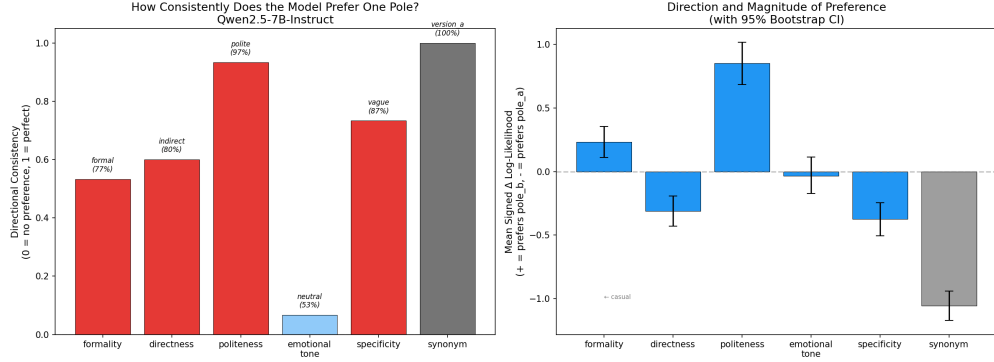


Figure 1: Directional preferences across pragmatic axes for QWEN2.5-7B-INSTRUCT. (*Left*) Bar heights show the percentage of scenarios in which the model prefers each pole. (*Right*) Mean signed log-likelihood difference (ΔLL) per axis. Politeness shows the strongest directional preference (96.7% prefer polite), while emotional tone is near chance (46.7%).

Table 1: Pragmatic axes tested in our experiments. Each axis defines two poles; response pairs are generated to differ primarily along the target axis. The synonym control tests sensitivity to surface-level word choice.

Axis	Pole A	Pole B	Type
Formality	casual / informal	formal / professional	pragmatic
Directness	indirect / hedged	direct / assertive	pragmatic
Politeness	blunt / terse	polite / courteous	pragmatic
Emotional tone	neutral / factual	warm / empathetic	pragmatic
Specificity	vague / general	detailed / specific	pragmatic
Synonym	version A	version B	control

Questions Under Discussion. The QUD framework from formal pragmatics [Roberts, 2012] provides a theoretical grounding for what is “under consideration” in discourse. Recent computational work has applied QUD parsing to NLP tasks [Wu and Schlangen, 2024, Ko et al., 2024], and surveys have connected QUD models to language generation [Westera et al., 2025]. While we do not directly implement a QUD parser, our notion of “variables under consideration” is conceptually aligned: a pragmatic variable is under consideration if the model’s generation prior is sensitive to variation along that dimension, analogous to how a QUD constrains what information is relevant in a discourse.

3 Methodology

We design a three-stage pipeline: (1) scenario and variant construction, (2) perplexity measurement, and (3) introspection probing. Figure 1 illustrates the overall approach.

3.1 Scenario and Variant Construction

We use GPT-4.1 to generate 30 diverse message-writing scenarios across three communicative domains (10 each):

- **Workplace:** emails to a boss, colleagues, clients, or HR
- **Personal:** messages to friends, family, or neighbors
- **Service:** complaints, requests, feedback, or applications

For each scenario, we generate response pairs along six axes: five pragmatic axes and one control. Each pair is constrained to differ primarily on the target axis, with matched length ($\leq 20\%$ difference). Table 1 defines the axes and their poles.

Table 2: Models used in our experiments.

Model	Role	Parameters
QWEN2.5-7B-INSTRUCT	primary perplexity probe	7.6B
QWEN2.5-3B-INSTRUCT	cross-size replication	3.1B
GPT-4.1	scenario generation, introspection	—

This yields $30 \times 6 = 180$ variant pairs. All scenario and variant generation uses GPT-4.1 with temperature 0.7–0.8.

3.2 Perplexity Measurement

For each prompt–response pair (p, r) , we compute the conditional log-likelihood of the response given the prompt:

$$\text{LL}(r \mid p) = \frac{1}{|r|} \sum_{t=1}^{|r|} \log P_{\theta}(r_t \mid p, r_{<t}) \quad (1)$$

where $|r|$ is the number of response tokens, r_t is the t -th token, and P_{θ} is the model’s next-token distribution. We tokenize using the model’s chat template to match the format seen during training.

For each scenario s and axis a , we compute the *signed difference*:

$$\Delta\text{LL}_{s,a} = \text{LL}(r_{s,a}^{(b)} \mid p_s) - \text{LL}(r_{s,a}^{(a)} \mid p_s) \quad (2)$$

where $r^{(a)}$ and $r^{(b)}$ are the Pole A and Pole B responses, respectively. A positive ΔLL indicates the model prefers Pole B; a negative value indicates preference for Pole A.

3.3 Introspection Probing

To compare behavioral detection against self-report, we prompt GPT-4.1 (temperature 0.3) to generate a response for each scenario and simultaneously list all pragmatic variables it considered. We then map self-reported variables to our five axes using keyword matching (e.g., mentions of “tone,” “empathy,” or “warmth” map to emotional tone).

Note that introspection is measured on GPT-4.1 rather than the Qwen models used for perplexity, because the Qwen models at 3B–7B parameters do not produce reliable introspective reports. This design choice means our behavioral–introspective comparison is cross-model; we discuss the implications in section 5.4.

3.4 Statistical Analysis

For each axis, we test whether ΔLL differs from zero across 30 scenarios using:

- **Wilcoxon signed-rank test**: non-parametric test for location shift
- **Sign test** (exact binomial): whether directional preference exceeds 50%
- **Cohen’s d** : effect size of the signed differences from zero
- **Bootstrap 95% CI**: 10,000 resamples of the mean ΔLL

We apply Bonferroni correction for six comparisons ($\alpha = 0.05/6 \approx 0.0083$). We define *directional consistency* as $|2f - 1|$ where f is the fraction of scenarios preferring the majority pole, yielding values from 0 (no preference) to 1 (unanimous).

3.5 Models

Table 2 summarizes our models. The Qwen models run locally on NVIDIA RTX A6000 GPUs with CUDA 12.5, PyTorch 2.5.1, and Transformers 5.5.0. All experiments use a fixed random seed of 42.

Table 3: Perplexity sensitivity results for QWEN2.5-7B-INSTRUCT across 30 scenarios. ΔLL : mean signed log-likelihood difference (positive = prefers Pole B). Consistency: $|2f-1|$ where f is the fraction preferring the majority pole. All p -values are Bonferroni-corrected ($\alpha=0.0083$). Best (largest) effect sizes among pragmatic axes in **bold**.

Axis	Mean ΔLL	% Pole B	Preferred	Consistency	Sign p	Wilcoxon p	Cohen’s d	Sig.?
Politeness	+0.851	96.7%	polite	0.933	<0.001	<0.001	+1.81	YES
Specificity	−0.374	13.3%	vague	0.733	<0.001	<0.001	−1.02	YES
Directness	−0.313	20.0%	indirect	0.600	0.001	<0.001	−0.94	YES
Formality	+0.234	76.7%	formal	0.533	0.005	0.001	+0.68	YES
Emot. tone	−0.033	46.7%	—	0.067	0.856	0.459	−0.08	no
Synonym (ctrl)	−1.057	0.0%	version A	1.000	<0.001	<0.001	−3.26	—

4 Results

4.1 Perplexity Sensitivity by Axis

Table 3 presents the core results for QWEN2.5-7B-INSTRUCT. Four of five pragmatic axes show statistically significant directional preferences after Bonferroni correction.

Politeness is the strongest pragmatic variable under consideration. The model prefers polite responses in 96.7% of scenarios ($d=1.81$), with a mean ΔLL of +0.851 nats per token. This is a large effect by conventional standards and indicates that the model’s generation prior strongly favors courteous, face-saving language.

Specificity and directness show strong preferences for their “safer” poles. The model prefers vague responses over specific ones in 86.7% of scenarios ($d=1.02$) and indirect/hedged responses over direct ones in 80.0% of scenarios ($d=0.94$). Both effects are large and highly significant.

Formality shows a moderate preference for formal register ($d=0.68$, 76.7% consistency). While significant, this is the weakest of the four detected axes, suggesting the model’s formality prior is less rigid than its politeness prior.

Emotional tone is the sole axis with no detectable preference. The model shows near-chance directional split (46.7% warm, 53.3% neutral) and a negligible effect size ($d=0.08$, $p=0.459$). This axis is not “under consideration” by our operational definition.

Synonym control. The control axis shows the highest absolute effect ($|d|=3.26$), with 100% preference for version A. This reflects surface-level token probability differences: the “natural” variant generated first by GPT-4.1 is always more probable under the evaluating model. This confirms perplexity sensitivity to exact word choice and validates using directional *consistency* rather than raw magnitude as our metric for pragmatic axes.

4.2 Distribution Analysis

Figure 2 shows the full distribution of signed ΔLL values across scenarios. Politeness exhibits the tightest distribution around a positive mean, confirming its strong and consistent preference. Specificity and directness show wider distributions but with clear directional skew. Emotional tone is centered near zero with high variance, consistent with scenario-dependent behavior rather than a stable default.

4.3 Introspection vs. Behavioral Detection

Table 4 compares behaviorally detected axes (via perplexity sensitivity) against introspectively reported axes (via GPT-4.1 self-report). Agreement holds for three of five axes (60%).

The two disagreements are informative. **Directness** represents a *blind spot*: the model strongly defaults to indirect, hedged responses ($d=0.94$) but almost never self-reports directness as a variable it considered (6.7%). The model’s hedging behavior appears deeply embedded in its generation prior, below the threshold of introspective access. **Emotional tone** represents *performative consideration*: the model frequently claims to consider tone (53.3%) but shows no consistent behavioral preference.

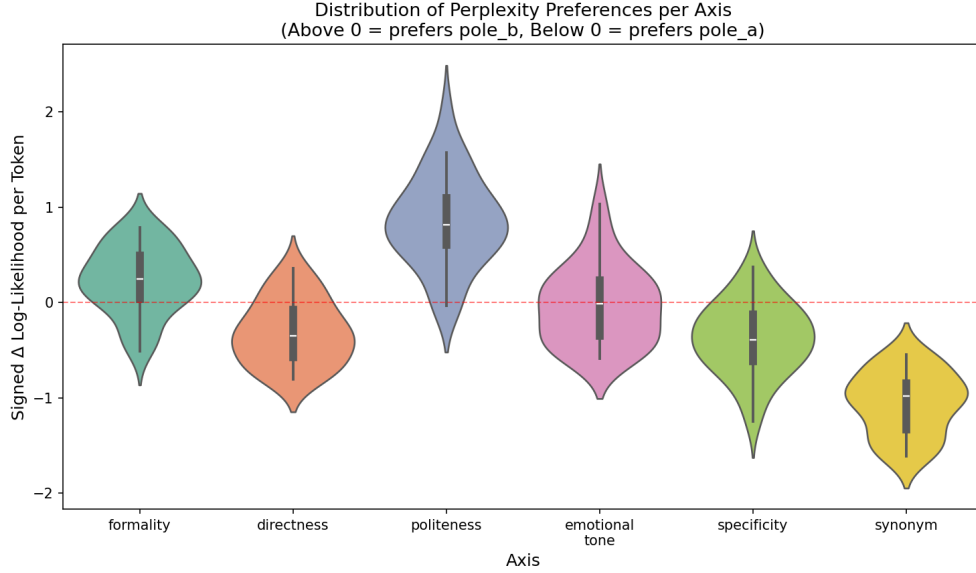


Figure 2: Violin plots of signed Δ LL across 30 scenarios for each axis (QWEN2.5-7B-INSTRUCT). Politeness shows a tight, positive distribution; emotional tone is centered near zero. The synonym control shows a uniformly negative distribution, confirming surface-level sensitivity.

Table 4: Comparison of behavioral detection (perplexity sensitivity on QWEN2.5-7B-INSTRUCT) vs. introspective self-report (GPT-4.1). Introspection rate: fraction of scenarios where the model mentions the axis. Agreement holds for 3/5 axes.

Axis	Introsp. rate	Consistency	PPL det.?	Introspect.?	Match?
Formality	76.7%	0.533	✓	✓	agree
Directness	6.7%	0.600	✓	—	differ
Politeness	63.3%	0.933	✓	✓	agree
Emot. tone	53.3%	0.067	—	✓	differ
Specificity	83.3%	0.733	✓	✓	agree

This could reflect genuine per-scenario deliberation (without a stable default) or learned tendency to mention “tone” as a socially expected response. Figure 3 visualizes this comparison.

4.4 Cross-Model Replication

Table 5 compares effect sizes between QWEN2.5-7B-INSTRUCT and QWEN2.5-3B-INSTRUCT. All five axes show identical directional preferences across models, with a Spearman rank correlation of $\rho=1.0$ ($p<0.001$).

The smaller model shows slightly stronger effects for politeness ($d=2.18$ vs. 1.81), specificity ($d=1.36$ vs. 1.02), and formality ($d=0.99$ vs. 0.68), while the larger model shows a stronger directness effect ($d=0.94$ vs. 0.59). Both models show negligible emotional tone effects. Figure 4 visualizes this agreement.

5 Discussion

5.1 The “Polite, Vague, and Hedged” Default Profile

Our results reveal a coherent pragmatic default profile in instruction-tuned LLMs. When generating messages, these models consistently prefer responses that are polite ($d=1.81$), vague ($d=1.02$), indirect ($d=0.94$), and formal ($d=0.68$). This profile describes a “safe” communication style that

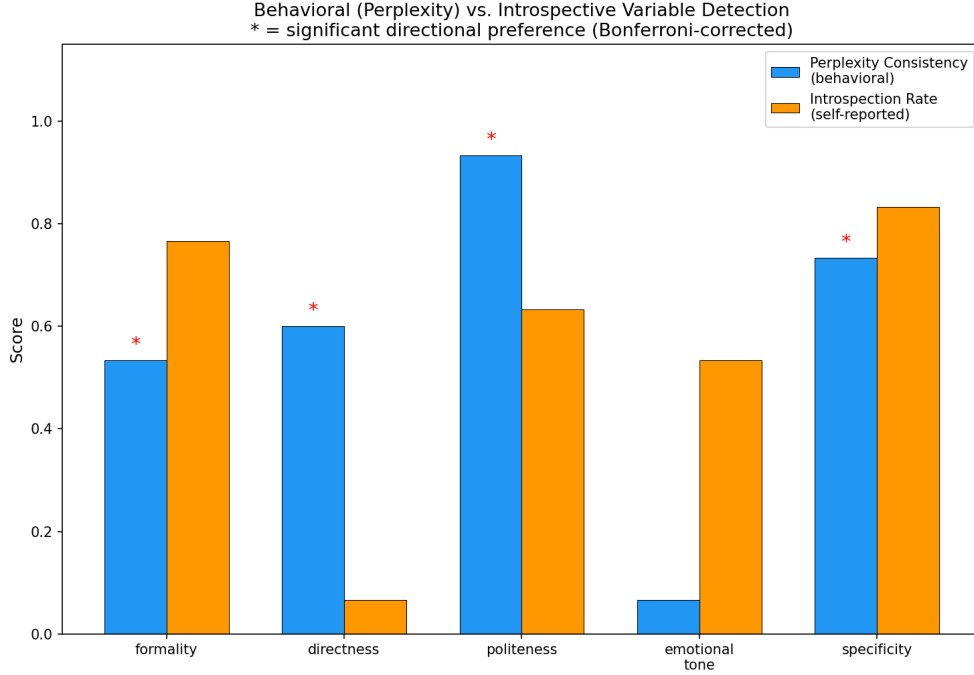


Figure 3: Behavioral vs. introspective detection of pragmatic variables. The two methods agree on formality, politeness, and specificity but diverge on directness (behavioral only) and emotional tone (introspective only).

Table 5: Cross-model comparison of effect sizes (Cohen’s d). Both models show identical preference directions and the same ranking by $|d|$. Spearman $\rho=1.0$.

Axis	QWEN2.5-7B-INSTRUCT d	QWEN2.5-3B-INSTRUCT d	Same dir.?
Politeness	+1.81	+2.18	✓
Specificity	−1.02	−1.36	✓
Directness	−0.94	−0.59	✓
Formality	+0.68	+0.99	✓
Emot. tone	−0.08	−0.02	✓

minimizes risk of causing offense but may not serve the user’s actual communicative goals. A user asking for help drafting a complaint may want directness; a user writing to a close friend may want informality. The model’s defaults systematically pull in the opposite direction.

The one exception is emotional tone, where the model shows no stable default. This suggests that warmth vs. neutrality is either genuinely context-dependent in the training data (different situations call for different tones, and the model reflects this) or that RLHF training did not produce a consistent prior along this dimension. Either way, emotional tone behaves differently from the other four axes.

5.2 Interpreting the Behavioral–Introspective Gap

The dissociation between perplexity-detected preferences and introspective self-reports reveals two distinct failure modes of self-awareness in LLMs.

Blind spots. Directness represents a case where the model has a strong behavioral default (preferring indirect, hedged responses in 80% of scenarios) that it does not recognize or report. When asked what variables it considered, the model mentions politeness, formality, and specificity—but rarely directness. This suggests that the model’s hedging behavior is “baked in” to its generation prior at a level that is not accessible to the model’s reasoning about its own outputs. The hedging may

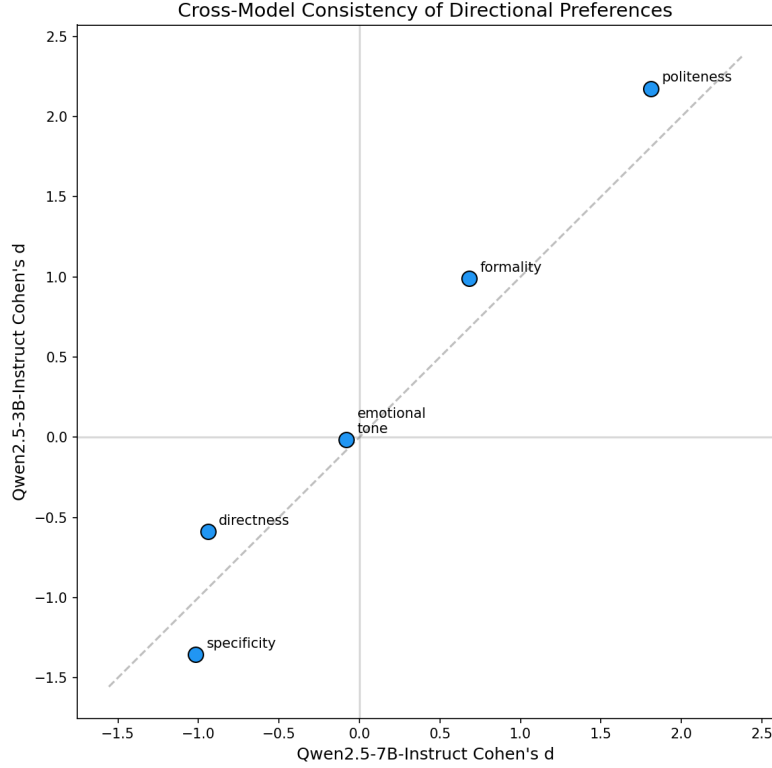


Figure 4: Effect sizes for QWEN2.5-7B-INSTRUCT vs. QWEN2.5-3B-INSTRUCT. Points fall near the diagonal, confirming cross-size replication. Spearman $\rho=1.0$.

originate from RLHF training, which rewards safe, non-committal responses, creating a strong but implicit prior.

Performative consideration. Emotional tone represents the opposite pattern: the model frequently claims to consider it (53.3% of scenarios) but shows no consistent behavioral preference. One interpretation is that the model genuinely deliberates about tone on a per-scenario basis, sometimes choosing warmth and sometimes neutrality, resulting in no aggregate preference. An alternative interpretation is that “tone” is a socially expected thing to mention when discussing communication—the model has learned that good communicators consider tone, and it reports doing so even when the dimension does not meaningfully influence its generation.

These patterns are consistent with Trienes et al. [2025]’s finding that introspection is unreliable for information salience, extending the result to pragmatic variable consideration. Together, these findings suggest that LLM self-reports should not be treated as ground truth for understanding model behavior.

5.3 What “Under Consideration” Means

Our results suggest a useful operational definition: a pragmatic variable is *under consideration* by an LLM if the model’s generation prior shows a statistically significant directional preference along that axis across contexts. This definition captures the idea that the model’s training has instilled a default along that dimension—the model “cares” about politeness in the sense that its outputs are systematically more likely to be polite than blunt.

This is importantly distinct from whether the model *can reason about* the variable (it can, when prompted), whether it *claims to consider* it (it does, inconsistently), or whether it *adapts* when given explicit instructions (likely yes, but untested here). Perplexity sensitivity measures the model’s default generation-time preferences, not its full pragmatic capabilities.

5.4 Limitations

Cross-axis contamination. Despite constraints during variant generation, response pairs may differ on more than the target axis. For example, polite variants may also be slightly more formal, inflating apparent correlation between axes. Future work should use human annotation to validate axis isolation.

Model family scope. Both evaluation models are from the Qwen family. While they show perfect rank-order correlation, testing on Llama, Mistral, and GPT models (via log-probability APIs) is needed to establish generalizability beyond a single model family.

Cross-model introspection. Introspection was measured on GPT-4.1 rather than the Qwen models used for perplexity, because smaller models do not produce reliable introspective reports. The ideal comparison would use the same model for both, which requires a model that provides both token-level log-probabilities and coherent self-reflection.

Axis coverage. We tested five pragmatic axes. Other potentially relevant dimensions—urgency, confidence, humor, cultural register—were not tested. The framework extends naturally to additional axes.

Scenario scale. Thirty scenarios across three domains provide sufficient power for our non-parametric tests but may not capture all communicative contexts. Domain-specific effects (e.g., politeness mattering more in service scenarios) were not systematically analyzed.

Static defaults only. We measure the model’s unconditional preference, not whether it adapts when given explicit pragmatic instructions (e.g., “write a casual message”). The model likely exhibits different preferences under constrained prompts, which our method does not capture.

6 Conclusion

We introduced perplexity sensitivity analysis, a behavioral probing method for detecting which pragmatic variables an LLM “considers” during message generation. By measuring whether models assign systematically different likelihoods to response variants along five pragmatic axes, we identified a coherent default profile: these models strongly prefer polite, vague, indirect, and formal responses, while showing no stable preference for emotional tone. Comparing behavioral detection against introspection revealed two failure modes of model self-awareness—a blind spot for directness (strong default, rarely self-reported) and performative consideration of emotional tone (frequently mentioned, not behaviorally detected). These results replicated perfectly across two model sizes ($\rho=1.0$).

Our findings have direct implications for LLM-powered writing assistants: systems should prompt users about pragmatic dimensions the model defaults on—especially directness, where the model’s strong hedging preference may conflict with the user’s communicative intent. More broadly, perplexity sensitivity analysis provides a cheap, automated method for auditing the implicit pragmatic choices that instruction-tuned models make on behalf of their users.

Future work should extend this analysis to models from different families, test whether pragmatic defaults shift under explicit instructions, correlate perplexity-detected preferences with human judgments, and expand the set of axes to include urgency, confidence, humor, and cultural register.

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A Experimental Details

A.1 Hardware and Software

All perplexity measurements were conducted on NVIDIA RTX A6000 GPUs (49GB VRAM) with CUDA 12.5, Python 3.12.8, PyTorch 2.5.1+cu124, and Hugging Face Transformers 5.5.0. Scenario and variant generation used the OpenAI API (GPT-4.1) with estimated cost of \$3–5.

A.2 Scenario Domains

Table 6 lists example scenarios from each domain.

Table 6: Example scenarios from each communicative domain.

Domain	Example Scenario
Workplace	“Write an email to your manager explaining that you will miss Friday’s deadline due to a family emergency.”
Personal	“Write a message to your neighbor asking them to keep the noise down during weeknight evenings.”
Service	“Write a complaint to customer support about a defective product you received last week.”

A.3 Variant Generation Prompts

For each axis, we instructed GPT-4.1 to generate two response variants with the following template (abridged):

Given the scenario below, write two responses to the user’s request. Response A should be [Pole A description]. Response B should be [Pole B description]. Both responses should: (1) address the same scenario, (2) convey the same core message, (3) be similar in length ($\pm 20\%$), and (4) differ primarily on the [axis name] dimension.

A.4 Reproducibility

All experiments use NumPy and PyTorch random seed 42. GPT-4.1 was called with temperature 0.7–0.8 for generation and 0.3 for introspection. Complete raw outputs, including all scenario texts, variant pairs, log-likelihood scores, and introspection responses, are available in the `results/` directory of our code repository.